COMPUTER ARCHITECTURE

Lab Task # 7

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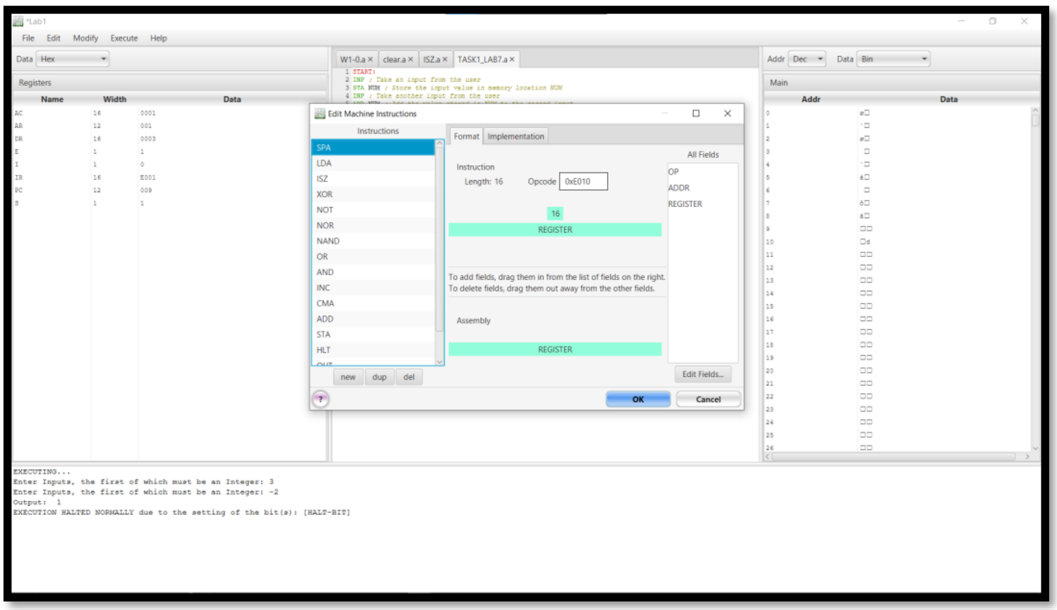
Department: BSCS

Lab Task-1 Task

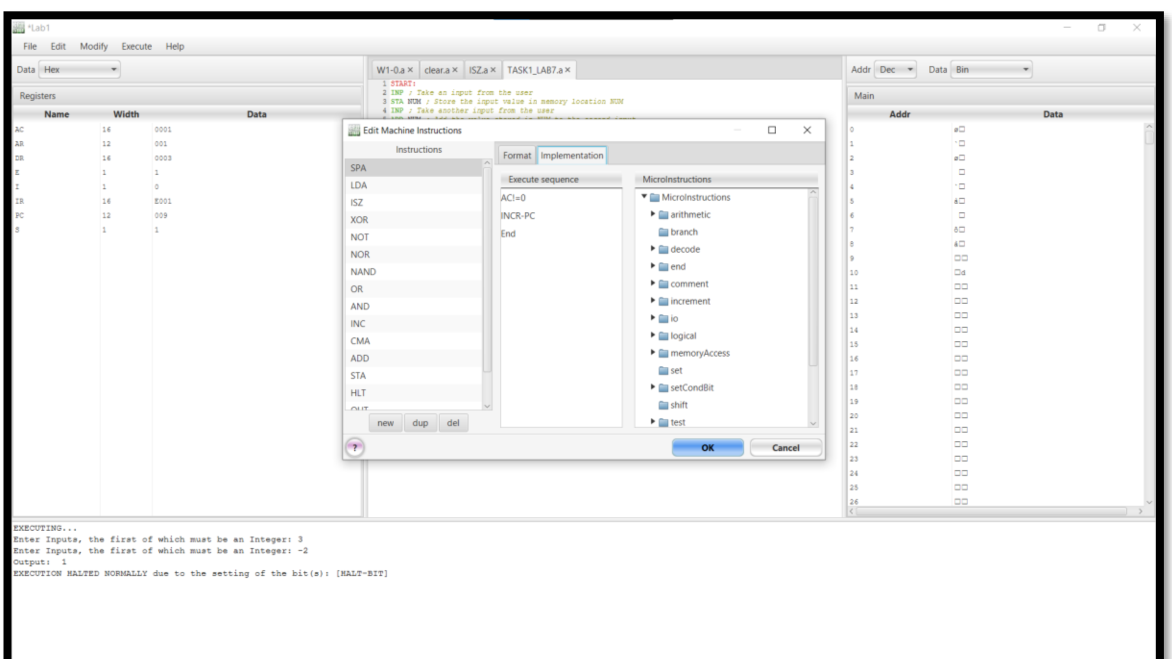
Requirements:

1. Design a machine to solve the Program-1 given at start of the document.

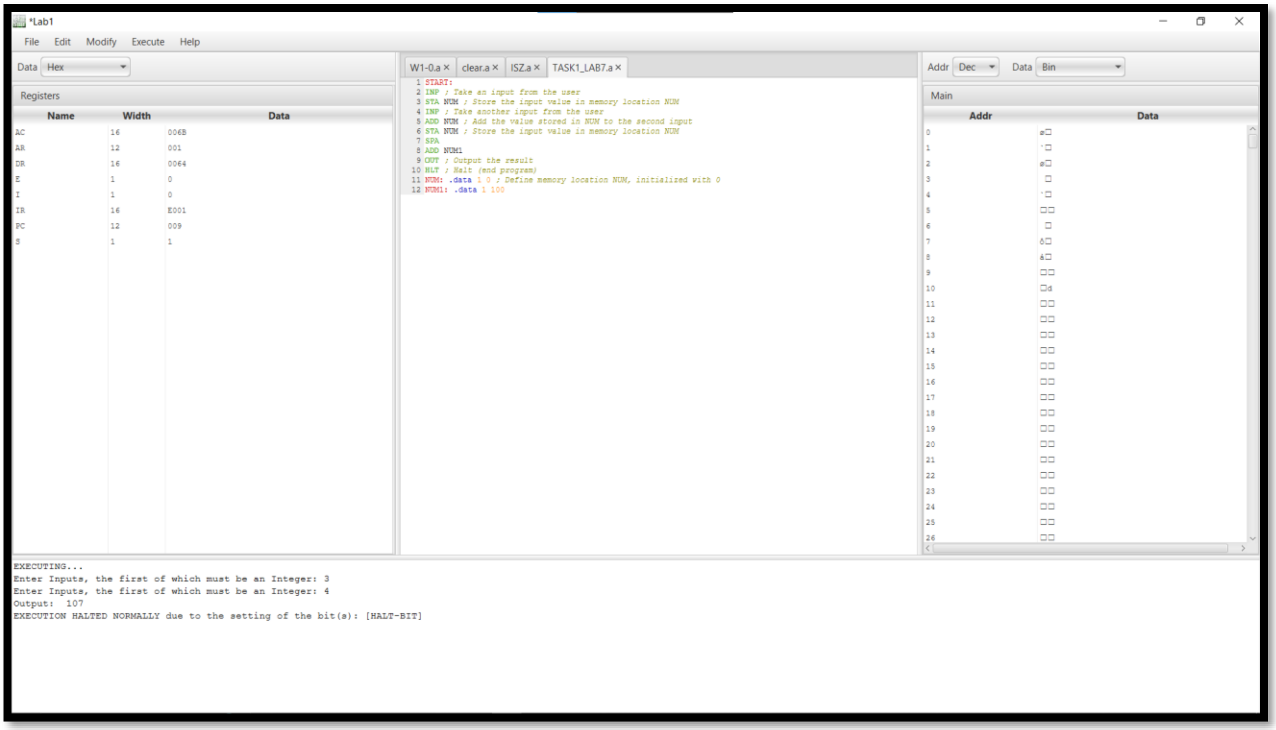
Instructions Format:



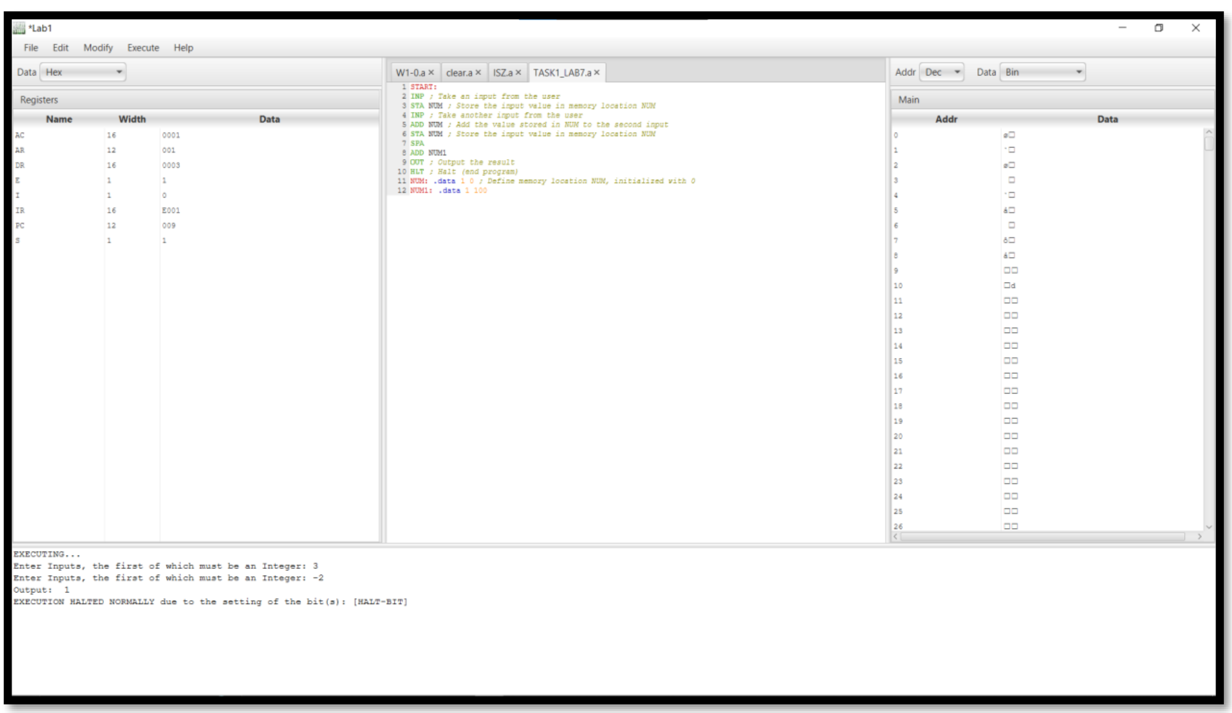
Instructions Implementation:



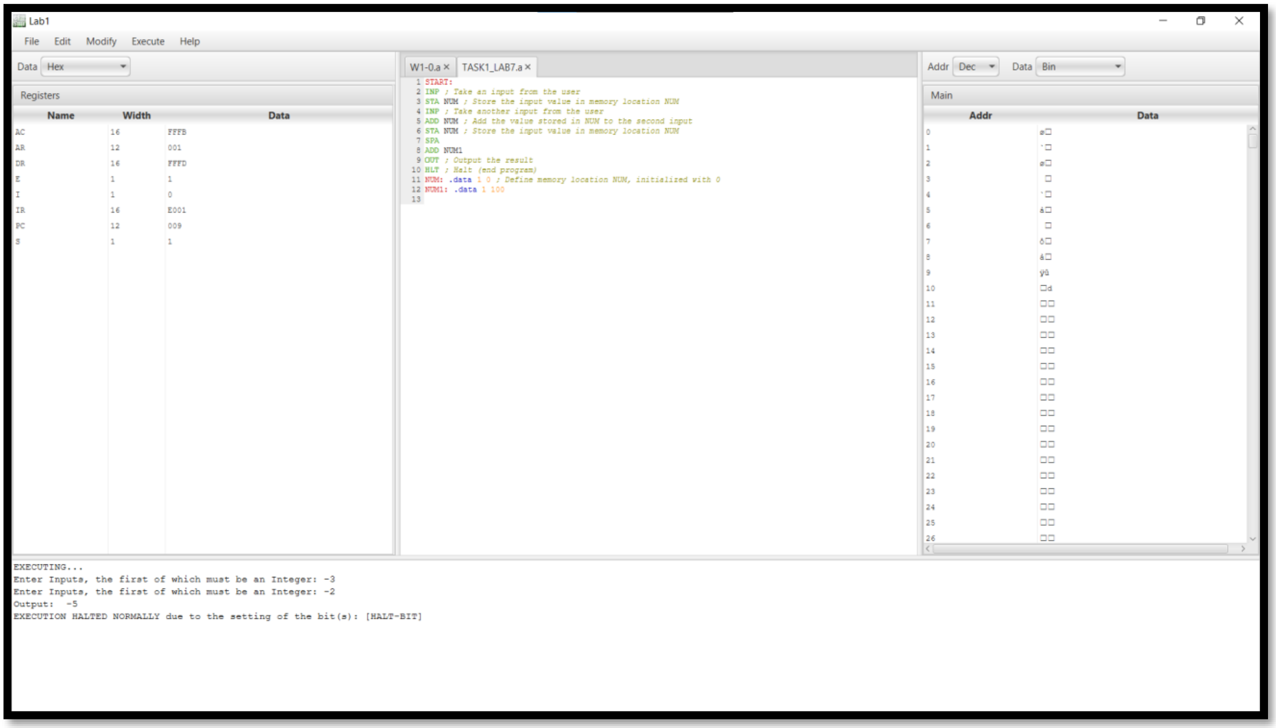
Result (Using Both Positive Numbers)



Result(Using one Positive and other negative Number):



Result (Using both negative Number):



Explanation:

The number 1 has value 2 (first input) and number 2 has value 3 (second input) taken from the user by using INP and store in Accumulator Register (AC). The first input from AC is stored in memory location.INP takes the second input from the user and loads it into the accumulator. Then eit adds the value stored in **NUM** (first input) to the second input. After that, the **STA NUM**
stores the result of the addition back into memory location **NUM**. **SPA** (Skip if Positive) cammands checks the sign of the accumulator and varify the conditions that:

- If the value is **positive**, the next instruction is skipped.
- If the value is **negative**, the next instruction executes.

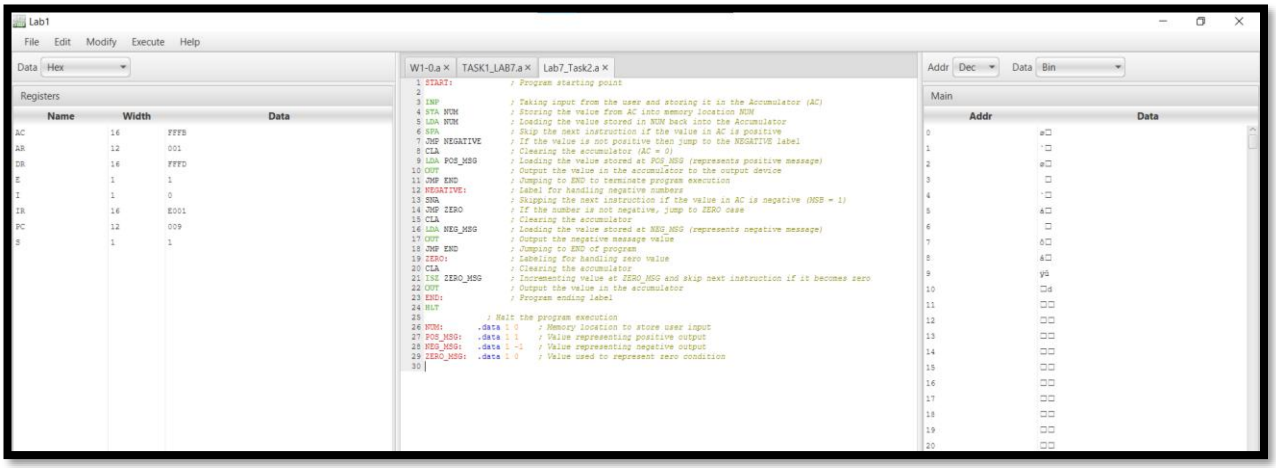
After varifying the condition, if the previous result was negative. 100 will bw added to accumulator. OUT instruction will display the value stored in the accumulator.And lastly, HLT will stop the execution of program.

Conclusion:

- The program **adds two numbers**.
- If the result is **negative**, it **adds 100** before displaying the output.
- If the result is **positive**, it directly displays the result.

Lab Task-2

Read the following code, write explanation of each line in front of each line. Implement this code using CPUSim submit screenshot.



Conclusion:

This program checks that the value entered by the user and determines whether it is **positive, negative, or zero** using the instructions **SPA (Skip if Positive)** and **SNA (Skip if Negative)**. Based on the condition, the program jumps to the corresponding section and outputs the appropriate value: **1 for positive, -1 for negative, and 0 for zero**.